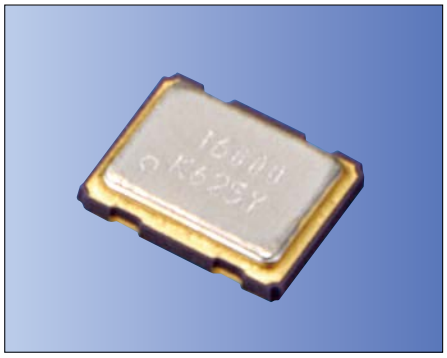
3.2 × 2.5mm for Automotive



AEC-Q200 PSL: R4Y

RoHS Compliant MSL1

Features

- Crystal unit for automotive electronics
- Improved solderability
- Improved mounting stability with 4 terminals
- Improved anti-noise performance with GND terminal
- Ceramic package • Miniature and low profile
- Improved rust prevention performance
- Reflow compatible
- Highly reliable solder junction (3000 heat cycles − 40 to +125°C)

Applications

- Automotive (ECU, High-Speed Automotive Network, TPMS)

How to Order

CX3225SA 25000 □□ □ □ □ □ HH
① ② ③④ ⑤ ⑥ ⑦ ⑧

Packaging
(Tape & Reel 3000 pcs./ reel)

①Series (Type and Size)
②Frequency (8,000 to 54,000kHz)
③④Load Capacitance (Unit:pF) Ex.: D1:8.1pF

③Integer part	
D	8
E	9
F	10
G	11
H	12
Z	Special suffix

④After the decimal point	
0	.0
1	.1
2	.2
3	.3
4	.4
5	.5
6	.6
7	.7
8	.8
9	.9

⑤Frequency Tolerance

F	±10×10⁻⁶
G	±15×10⁻⁶
H	±20×10⁻⁶
J	±25×10⁻⁶
K	±30×10⁻⁶
M	±40×10⁻⁶
P	±50×10⁻⁶
W	All Over
Z	Special suffix

⑥Operating Temp. Range ⑦Frequency Temp. Stability
Note: Supported for frequencies of 12MHz or higher. For frequencies ranging from 8MHz to less than 12MHz, please contact us.

⑥ \ ⑦		J ±15ppm	L ±20ppm	Q ±30ppm	R ±40ppm	S ±50ppm	T ±100ppm	V ±150ppm	W ±200ppm
L	-30 to 85°C	✓	✓	✓	✓	✓	✓	✓	✓
P	-40 to 85°C		✓	✓	✓	✓	✓	✓	✓
R	-40 to 105°C			✓	✓	✓	✓	✓	✓
S	-40 to 125°C				✓	✓	✓	✓	✓
T	-40 to 150°C						✓	✓	✓

⑧Individual Specification (STD Specification is “HH” .)

Specifications

Item	Symbol	Specification	Unit	Remarks
Frequency Range	f_{nom}	8000 to 54000	kHz	Please contact us for other frequency range.
Overtone Order	OT	Fundamental	—	
Load Capacitance	CL	8	pF	Please contact us for other CL requirements.
Frequency Tolerance	f_{tol}	±15	×10⁻⁶	25°C± 3°C
Motional Series Resistance	R1	Table 1	ohm	
Drive Level	DL	10μW(200μW max.)	μ W	
Operating Temp. Range	T_{use}	− 40 to +150	°C	
Storage Temp. Range	T_{stg}	− 40 to +150	°C	
Frequency Temp. Characteristics	f_{tem}	± 150	× 10⁻⁶	Freq. deviation from the value at 25°C
Shunt capacitance	C0	3.0max.	pF	

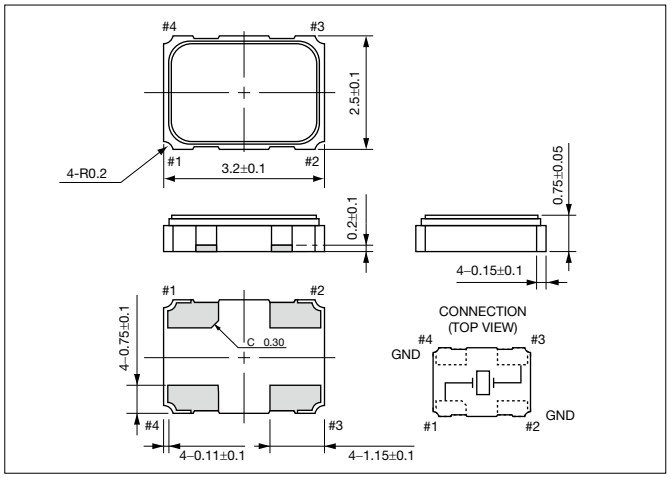
Please contact us for other specifications.

Table 1 Motional Series Resistance ESR (CI) STD Specification

Frequency (MHz)	ESR (Ohm)
8≦F<9.8	500
9.8≦F<12	200
12≦F<16	120
16≦F<20	60
20≦F≦54	50

Dimensions

(Unit: mm)



Recommended Land Pattern

(Unit: mm)

